

Name: _____

Period: _____

Date: _____

Topic 1.5 Guided Notes: Sleep

Unit 1: Biological Bases of Behavior | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appspsychreviewhub.com/overviews/topic1.5

Part 1: Consciousness and Our Internal Clock

Word Bank

<i>Shift Work</i>	<i>Jet Lag</i>	<i>Consciousness</i>
<i>Circadian Rhythm</i>	<i>Sleep</i>	

- _____: a person's awareness of themselves and their environment, including thoughts, feelings, sensations, and perceptions at any given moment. It exists in varying levels throughout the day, not as a simple on/off switch.
- _____: a naturally recurring state of rest characterized by reduced consciousness, slowed bodily functions, and brain activity that supports restoration, memory consolidation, and overall health.
- _____: the internal biological clock that regulates the sleep-wake cycle and other physiological processes on a roughly 24-hour schedule.
- _____: fatigue and sleep disturbance resulting from the disruption of the body's normal circadian rhythm following long flights through different time zones.
- _____: schedules that take place outside the traditional 9-to-5 day, which can severely disrupt circadian rhythms and sleep patterns, heavily impacting how alert you feel when you are awake.

Part 2: Riding the Waves (The Stages of Sleep)

Word Bank

<i>Alpha Waves</i>	<i>NREM Stage 1</i>	<i>REM Rebound</i>
<i>Delta Waves</i>	<i>NREM Stage 2</i>	<i>REM Sleep</i>
<i>Sleep Spindles</i>	<i>NREM Stage 3</i>	<i>Hypnagogic Sensations</i>
<i>Theta Waves</i>		

When you drift off, your brain doesn't just turn off. Instead, it moves through a series of predictable stages, each identified by its own unique EEG pattern. NREM stages actually decrease in duration as successive cycles repeat throughout the night.

- _____: relatively slow brain waves your brain produces when you are awake but relaxed and calm, often occurring just before falling asleep.
- _____: a very light sleep stage marked by slowed brain activity, muscle relaxation, and slow _____. During this brief window, you may experience _____ — vivid, dreamlike experiences such as falling or seeing flashes of light — which is exactly what causes that sudden physical jerk that snaps you back awake.
- _____: a deeper stage of light sleep characterized by a slower heart rate and body temperature. Brain activity displays _____, brief bursts of rapid brain activity believed to play a crucial role in memory consolidation and protecting your sleep from disturbances.
- _____: the deepest stage of non-REM sleep, defined by the presence of very slow, high-amplitude _____. This stage is absolutely vital for physical restoration, growth, and recovery.
- _____: an active stage marked by rapid eye movements, vivid dreaming, high brain activity, and temporary muscle paralysis. It's often considered paradoxical because its EEG shows active brain waves similar to wakefulness, yet the skeletal body remains completely relaxed and immobilized.
- _____: an increase in REM sleep that occurs after a person is deprived of it, causing the body to spend much more time in REM during subsequent sleep cycles to make up for the loss.

Apply It: Which Sleep Stage?

Using the word bank above, identify which sleep stage or wave type best matches each scenario.

#	Scenario	Stage/Wave
1	An EEG shows slow, high-amplitude delta waves as a child sleeps soundly through her most physically restorative stage of the night.	
2	A sleeping teenager's EEG shows brief bursts of rapid brain activity between longer stretches of slower waves — a pattern linked to protecting her sleep from the noise of her sibling's music down the hall.	
3	A woman's eyes dart rapidly beneath her eyelids while the rest of her body lies completely still and limp.	
4	A man's brain waves slow into a relaxed, drowsy pattern as he lies in a hammock with his eyes closed, not yet fully asleep.	

Part 3: Why Do We Sleep and Dream?

Word Bank

Restoration Theory of Sleep

Consolidation Theory

Activation-Synthesis Theory

According to the _____, our nightly downtime serves to repair and rejuvenate the body and brain, helping restore energy, support growth, and maintain overall physical and mental functioning.

Alternatively, the _____ focuses on cognition, proposing that sleep — especially REM and deep NREM sleep — helps stabilize and strengthen newly formed memories so they can be stored long-term.

When it comes to the structure of dreams, the _____ suggests that dreams are simply the brain's attempt to make sense of random neural activity occurring during sleep — the brainstem fires random signals during REM, and your higher brain regions synthesize those electrical sparks into a cohesive story.

What's on the Exam? Psychoanalytic dream theories (like Sigmund Freud's ideas on hidden meanings, latent content, and manifest content) are completely outside the scope of the exam. Keep your focus on biological and cognitive perspectives like activation-synthesis and consolidation!



Video Notes: To Sleep, Perchance to Dream (Crash Course Psychology #9)

Answer the questions below as you watch — some ideas will already be familiar from Parts 1 and 2, and that's fine; reinforcing them is the point.

- Who were Aserinsky and Kleitman, and what tool did they use to help discover REM sleep?
- What are the four stages of sleep described in the video, in order?
- Why does the video call REM sleep "paradoxical"?
- About how often does a full sleep cycle (NREM 1, 2, 3, and REM) repeat throughout the night, according to the video?
- According to the video, what are some negative effects of sleep deprivation on the body and mind?
- According to the video, why might everyday things (like a car alarm or a game you played earlier) show up in your dreams?

Part 4: Disruptions of the Night (Sleep Disorders)

Word Bank

<i>Insomnia</i>	<i>Narcolepsy</i>	<i>Night Terrors</i>
<i>Somnambulism</i>	<i>REM Sleep Behavior Disorder</i>	<i>Sleep Apnea</i>

- _____: a common sleep disorder characterized by persistent difficulty falling asleep, staying asleep, or getting restful sleep despite having adequate opportunity to do so.
- _____: sudden, uncontrollable sleep attacks during waking hours, often including rapid entry into REM sleep and a possible loss of muscle control.
- _____: a dangerous condition marked by temporary cessations of breathing during sleep and repeated momentary awakenings, which often leaves individuals completely exhausted during the day.
- _____: occurs when the normal muscle paralysis of REM sleep is absent, causing a person to physically act out their dreams, sometimes violently.
- _____ (sleepwalking): a disorder in which a person walks or performs other complex behaviors while still asleep, usually during deep NREM Stage 3 sleep, with little or no memory of the event.
- _____: sudden episodes of intense fear, screaming, and high arousal during deep NREM sleep, typically leaving little to no recall of the event upon waking.

Apply It: Which Sleep Disorder?

#	Scenario	Disorder
1	A man's arms and legs act out a chase scene from his dream, kicking and swinging at his sleeping wife beside him.	
2	A teenager suddenly collapses into deep sleep in the middle of gym class despite a full night's rest.	
3	A young boy is found calmly opening the refrigerator at 2 a.m. with his eyes open, and has no memory of it the next morning.	
4	A woman lies awake for hours most nights, unable to fall asleep despite feeling exhausted.	

5. Don't Trip Up! (Common Misconceptions)

_____ are vivid, scary dreams that occur during REM sleep, meaning you can usually remember them when you wake up. _____ occur during deep NREM Stage 3 sleep, involve intense physical panic (like screaming or sweating), and leave almost no memory trace behind.

_____ happens during deep NREM Stage 3 sleep when the motor cortex is active but dreams aren't actively happening. Physically acting out a vivid dream because of a failure in protective muscle paralysis is specifically classified as _____.

6. Quick Check: Review Questions

Circle the best answer for each question below.

1. While drifting off to sleep, a man suddenly experiences a vivid sensation of falling and his body jerks, snapping him back awake. This experience is best described as a:
 - A. REM Rebound
 - B. Sleep Spindle
 - C. Hypnagogic Sensation
 - D. Night Terror
2. An EEG recording of a sleeping participant shows brief bursts of rapid brain activity believed to play a role in memory consolidation and protecting sleep from outside disturbances. This pattern indicates the participant is in:
 - A. NREM Stage 3
 - B. REM Sleep
 - C. NREM Stage 2
 - D. NREM Stage 1
3. During a specific stage of sleep, a sleeper's EEG shows fast, active brain waves nearly identical to being awake, yet his skeletal muscles are completely paralyzed and unable to move. This seemingly contradictory combination is why this stage is often called:
 - A. NREM Stage 1
 - B. REM Sleep
 - C. The Restoration Stage
 - D. NREM Stage 2
4. A psychologist explains that dreams are not hidden messages from the unconscious mind, but are instead simply the brain's attempt to make sense of random neural signals fired by the brainstem during REM sleep. This explanation reflects the:
 - A. Activation-Synthesis Theory
 - B. Consolidation Theory
 - C. Freud's Psychoanalytic Theory
 - D. Restoration Theory of Sleep
5. A man's wife reports that he repeatedly stops breathing for several seconds throughout the night, followed by a gasping momentary awakening, leaving him exhausted despite spending eight hours in bed. Which disorder is most consistent with this description?
 - A. Insomnia
 - B. Narcolepsy
 - C. Night Terrors
 - D. Sleep Apnea

6. A young girl suddenly sits up screaming in intense fear during the deepest stage of her NREM sleep, but the next morning she has no memory of the episode at all. This is best classified as a:

- A. REM Rebound
- B. Night Terror
- C. Hypnagogic Sensation
- D. Nightmare



Check Your Progress

Done with your notes? Head over to the [Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>

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